EXERCISE-I (Conceptual Questions) Build Up Your Understanding

PROPERTIES AND COMPOUNDS

1. Correct order of density is –
(1) Li > Na (2) K > Na
(3) Mg > Ca (4) Cs < Rb
SB0001

2. Which is having highest m.p. –
(1) Be (2) Mg (3) Ca (4) Sr
SB0002

3. Weak reductant in alkali metal is –
(1) Li (2) Na (3) K (4) Cs
SB0003

4. The metal used in photoelectric cell is –
(1) Na (2) Cs (3) Mg (4) Ca
SB0004

5. Lithium chloride is highly soluble in –
(1) C_6H_6 (2) H_2O (3) D_2O (4) All
SB0005

6. Which metal will not form superoxide –
(1) Li (2) Be (3) Na (4) All
SB0006

7. More stable hydride is –
(1) Cs – H (2) Rb – H
(3) K – H (4) Li – H
SB0007

8. In which compound hydrogen is electronegative –
(1) CaH_2 (2) CH_4 (3) HCl (4) All
SB0008

9. Which of the following metal will give apple green colour on Bunsen flame –
(1) Ba (2) Sr
(3) Ca (4) K
SB0009

10. The density of –
(1) Na > K (2) Na = K
(3) K > Na (4) Li > K
SB0010

11. Ionic conductances of hydrated M^+ ions are in the order –
(1) $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) > Cs^+(aq)$
(2) $Li^+(aq) > Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$
(3) $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) < Cs^+(aq)$
(4) $Li^+(aq) < Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$
SB0012

12. Which of the following halides has the highest melting point –
(1) NaCl (2) KCl
(3) NaBr (4) NaF
SB0013

13. Which of the following does not give an oxide on heating –
(1) $MgCO_3$ (2) Li_2CO_3
(3) $ZnCO_3$ (4) K_2CO_3
SB0014

14. Molten potassium chloride conducts electricity due to the presence of –
(1) Free electron
(2) Free ions
(3) Free molecules
(4) Atom of potassium & chloride
SB0017

15. Which of the following element have maximum tendency to form complex compound –
(1) Be (2) Ba (3) Ca (4) Mg
SB0018

16. On heating sodium metal in the current of dry ammonia leads to the formation of which gas–
(1) $NaNH_2$ (2) NaN_3
(3) NH_3 (4) H_2
SB0019

17. Sodium reacts with water more vigorously than lithium because it –
(1) Has higher atomic weight
(2) Is more electronegative
(3) Is more electropositve
(4) Is a metal
SB0020

18. The strongest reducing agent is –
(1) Be (2) Mg (3) Sr (4) Ba
SB0022

19. Both Be and Al become passive on reaction with conc. nitric acid due to –
(1) The non reactive nature of the metal
(2) The non reactive nature of the acid
(3) The formation of an inert oxide layer on the surface of the metals
(4) None of these
SB0023

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20. Sodium loses its lustre on exposure to moist air due to formation of –
(1) Na_2O, NaOH and Na_2CO_3
(2) Na_2O and NaOH
(3) Na_2O and Na_2CO_3
(4) NaOH and Na_2CO_3
SB0024

21. Potassium carbonate when heated to high temperature.
(1) Gives CO_2 (2) Gives O_2
(3) Gives CO (4) Gives no gas at all
SB0025

22. On Flame test K give -------- colour –
(1) Golden yellow (2) Crimson red
(3) Violet (4) Apple green
SB0026

23. An element having electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$ will form –
(1) Acidic oxide (2) Basic oxide
(3) Amphoteric oxide (4) Neutral oxide
SB0027

24. Which decomposes on heating –
(1) NaOH (2) KOH (3) LiOH (4) RbOH
SB0028

25. Which metal does not form ionic hydride –
(1) Na (2) Rb (3) Ca (4) Be
SB0029

26. The element of IA group which combines directly with nitrogen is –
(1) Li (2) Na (3) K (4) Cs
SB0030

27. Which of the following is paramagnetic
(1) K_2O (2) K_2O_2 (3) KO_2 (4) Na_2O
SB0032

28. A compound which upon hydrolysis releases ammonia is –
(1) Li_3N (2) $LiNO_3$
(3) $NaNO_3$ (4) None of these
SB0033

29. Which statement will be true for solution, when Ba is dissolved in ammonia:-
(1) Solution becomes blue
(2) Solution becomes good conductor
(3) Solution remains colourless
(4) Both (1) and (2) are correct
SB0037

30. In K, Rb and Cs, the decreasing order of reducing power in gaseous state is:-
(1) K > Cs > Rb (2) Cs > Rb > K
(3) K < Cs < Rb (4) Rb > Cs > K
SB0038

31. The correct order of density of following elements is:- (Be, Mg, Ca, Sr)
(1) Be > Mg > Ca > Sr (2) Ca > Mg > Be > Sr
(3) Ca < Mg < Be < Sr (4) Mg < Ca < Sr < Be
SB0039

32. On addition of metal, colour of liquid NH_3 solutions converts into bronze, the reason is :-
(1) Ammoniated electrones
(2) Metal amide formation
(3) Liberation of NH_3 gas
(4) Cluster formation of metal ions
SB0041

33. On allowing ammonia solution of s-block metals to stand for a long time, blue colour becomes fade. The reason is:-
(1) Formation of NH_3 gas
(2) Formation of metal amide
(3) Cluster formation of metal ions
(4) Formation of metal nitrate
SB0042

34. Which of the following s-block element reacts with NaOH to give water soluble complex :-
(1) Al (2) Ca (3) Be (4) Li
SB0043

35. When Na and Li placed in dry air we get :-
(1) NaOH, Na_2O, Li_2O
(2) Na_2CO_3, Na_2O_2, Li_2O
(3) Na_2O, Li_3N, NH_3
(4) Na_2O, Li_2O, Li_3N
SB0045

36. Which of the following oxide having O_2^{-2} (peroxide) anion :-
(1) Na_2O (2) BaO_2 (3) RbO_2 (4) KO_2
SB0046

37. Generally which of the following properties of IA group metals increases as the atomic number rises:
(a) Metallic character (b) Ionic radius
(c) Melting point (d) Density
(e) Ionisation potential
Correct answer is :-
(1) a, b, c (2) a, b, d (3) c, d, e (4) All
SB0047

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38. Which of the following s-block metals do not impart any colour to the flame
(1) Li, Be (2) Cs, Fr (3) Be, Mg (4) Ba, Ra
SB0048

39. Which can not be used to generate H_2 :–
(1) Al + NaOH
(2) Zn + NaOH
(3) Mg + NaOH
(4) LiH + H_2O
SB0049

40. Only those elements of s-block can produce superoxides which have :-
(1) High ionisation energy
(2) High electronegativity
(3) High charge density
(4) Low ionisation potential
SB0050

41. Alkali metals dissolve in liquid NH_3 then which of the following observations is not true:
(1) It becomes paramagnetic
(2) Solution turns into blue due to solvated electrons
(3) It becomes diamagnetic
(4) Solution becomes conducting
SB0052

42. Alkali metals give colour in bunsen flame due to –
(1) Low electronegativity
(2) One e^- in outer most orbit
(3) Smaller atomic radii
(4) Low ionisation energy
SB0053

43. Which of the following ions forms a hydroxide that is highly soluble in water ?
(1) K^+ (2) Zn^{2+} (3) Ni^{2+} (4) Al^{3+}
SB0054

44. The plaster of paris is hardened by
(1) Liberating CO_2
(2) Giving out water
(3) Combining with water
(4) Changing into $CaCO_3$
SB0056

45. Which of the following alkali metal carbonate is the least stable and decomposes readily
(1) Li_2CO_3 (2) Na_2CO_3
(3) K_2CO_3 (4) Cs_2CO_3
SB0057

46. In the reaction $M + O_2 \longrightarrow MO_2$ (super oxide) the metal is
(1) Li (2) Na (3) K (4) Ba
SB0058

47. Li does not resemble other alkali metals in following properties
(1) Li_2CO_3 decomposes into oxides while other alkali carbonates are thermally stable
(2) LiCl is predominantly covalent
(3) Li_3N is stable
(4) All
SB0059

48. Be and Al resemble in
(1) Both become passive on reaction with HNO_3 due to formation of oxide layer
(2) Their chlorides are lewis acids
(3) Hydroxides are soluble in alkali as well as in acid
(4) All
SB0060

49. Consider the following points
(a) Cs is the strongest reducing agent in IA group element
(b) Be does not form peroxide in II A group elements
(c) The density of potassium is less than sodium
(d) In alkali metals Li, Na, K and Rb, lithium has the minimum value of M.P.
Point out that the statement -
(1) (a) & (b) are correct
(2) (a), (b) & (c) are correct
(3) (b) & (c) are correct
(4) (b), (c) & (d) are correct
SB0061

50. Mg^{+2} does not form either peroxide or superoxide, because
(1) Mg^{+2} ion is relatively bigger
(2) Mg^{+2} ion is relatively smaller
(3) Mg^{+2} ion is stable
(4) Mg^{+2} ion is unstable
SB0062

51. Which of the following is true about Alkali metals
(1) All form solid bicarbonates
(2) All form ionic salt like hydride MH
(3) All form superoxide like KO_2
(4) All form nitrides
SB0064

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52. In between the metals A and B both form oxide but only B forms nitride, when both burn in air so A and B are
(1) Cs, K (2) Mg, Ca
(3) Li, Na (4) K, Mg
SB0066

53. There is loss in weight when mixture of Li_2CO_3 and $Na_2CO_3.10H_2O$ is heated strongly. This loss is due to :
(1) Li_2CO_3 (2) $Na_2CO_3.10H_2O$
(3) both (4) none
SB0070

Note : Q.54 to 57 are based on following reaction (s) :

$A \xrightarrow{\Delta} B\text{ (oxide)} + CO_2$
$B + H_2O \longrightarrow C$
$C + CO_2 \longrightarrow A\text{ (milky)}$
$B + NH_4Cl \xrightarrow{\Delta} D\text{ (gas)}$
$D + H_2O + CO_2 \longrightarrow E$
$E + NaCl \longrightarrow F$
$F \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$

54. Name of the process is :
(1) solvay (2) ammonia-soda
(3) both correct (4) none is correct
SB0071

55. A is :
(1) $Ca(HCO_3)_2$ (2) $CaCO_3$
(3) CaO (4) Na_2CO_3
SB0072

56. B and C are :
(1) CaO, $Ca(OH)_2$ (2) $Ca(OH)_2$, $CaCO_3$
(3) $CaCO_3$, $Ca(OH)_2$ (4) $Ca(OH)_2$, CaO
SB0073

57. D, E and F are :
(1) NH_3, NH_4Cl, NH_4HCO_3
(2) NH_3, NH_4HCO_3, $NaHCO_3$
(3) NH_4HCO_3, Na_2CO_3, $NaHCO_3$
(4) None
SB0074

58. A wire of an alkaline earth metal X, burnt in air and dipped in water, a gas 'Y' is evolved X and Y are respectively :–
(1) Na, NO_2 (2) Be, NO_2
(3) Mg, CO_2 (4) Mg, NH_3
SB0075

59. Ratio of SiO_2 and Al_2O_3 for a good quality cement.
(1) 2.5 – 4.0 (2) 1.5 – 2.5
(3) 3.5 – 5.0 (4) 0.5 – 1.5
SB0112

60. To obtain pure NaCl final solution is saturated with which of the following gas :
(1) NH_3 (2) N_2 (3) O_2 (4) HCl
SB0113

61. Solvay process cannot be used for the manufacture of K_2CO_3 since
(1) $KHCO_3$ is having high lattice energy and Less solubility
(2) $KHCO_3$ is having high hydration enthalpy and more solubility
(3) $KHCO_3$ is having high MP
(4) All
SB0114

62. $BeO + C + Cl_2 \underset{}{\overset{(\Delta)}{\rightleftharpoons}} A + CO$
Correct statement(s) about 'A'
(1) It has sp^2 hybridisation of Be in its polymeric form @neetquestionpaper
(2) It has sp^2 hybridisation in its dimeric form
(3) Its dimeric structure is non planar
(4) It has $3C–2e^-$ bond in its polymeric structure
SB0115

63. Solubility of carbonate, sulphates, hydroxides of alkaline earth metals has an order
(1) increase, increase, increase
(2) decrease, increase, increase
(3) decrease, decrease, increase
(4) decrease, decrease, decrease
SB0116

64. Lime water turns milky by the passage of CO_2 and milkyness disappears when excess of CO_2 is passed Compounds responsible for these processes are :
(1) $Ca(HCO_3)_2$, $CaCO_3$ (2) $CaCO_3$, $Ca(HCO_3)_2$
(3) $CaCO_3$, CaO (4) $Ca(OH)_2$, $Ca(HCO_3)_2$
SB0117

65. Gypsum $\xrightarrow{393K}$ [A] $\xrightarrow{>393K}$ [B] Formula of A and B
(1) $CaSO_4$, $2(CaSO_4).H_2O$
(2) $CaSO_4.2H_2O$, $2(CaSO_4).H_2O$
(3) $2(CaSO_4).H_2O$, $CaSO_4$
(4) $CaSO_4$, $CaSO_4.\frac{1}{2}H_2O$
SB0118

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66. Sodium metal can be used for the drying of

(1) Ethanol (2) Phenol

(3) Ammonia (4) Dimethyl ether

SB0119

67. Which of the metal is used with aluminium to form air craft parts.

(1) K (2) Li

(3) Na (4) None

SB0120

68. Which of the following used as a important constituent of glass industry.

(1) $LiCO_3$ (2) Na_2CO_3

(3) $MgCO_3$ (4) None

SB0121

69. Which metal maintain ion balance and nerve impulse conduction in human body

(1) Na (2) K

(3) Both 1 and 2 (4) None

SB0122

70. _____ is used in manufacturing of artificial silk.

(1) NaOH (2) RbOH

(3) KOH (4) LiOH

SB0123

71. Which substance is used for making window of x-ray tubes.

(1) Beryllium (2) Strontium

(3) Barium (4) Radium

SB0124

72. Substance used in purification of sugar :

(1) $Ca(OH)_2$ (2) BeO

(3) $BeCl_2$ (4) CaO

SB0125

73. Select the correct statement regarding s-block element's -

(1) They are all are reactive metal with low ionisation energies

(2) The metallic charcter increases down the group

(3) They are never found pure in nature

(4) All of these

SB0126

74. Correct order of melting point is ?

(1) Be > Ca > Sr > Ba > Mg

(2) Ba > Sr > Ca > Mg > Be

(3) Be > Mg > Ca > Sr > Ba

(4) Be > Sr > Ca > Ba > Mg

SB0127

75. Indian salt peter is :-

(1) KNO_3 (2) $LiNO_3$ (3) $NaNO_3$ (4) None

SB0128

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EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**1**	**2**	**2**	**1**	**4**	**4**	**1**	**1**	**1**	**4**	**4**	**4**	**2**	**1**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**4**	**3**	**4**	**3**	**1**	**4**	**3**	**2**	**3**	**4**	**1**	**3**	**1**	**4**	**2**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**3**	**4**	**2**	**3**	**4**	**2**	**2**	**3**	**3**	**4**	**3**	**4**	**1**	**3**	**1**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**3**	**4**	**4**	**3**	**2**	**2**	**4**	**3**	**3**	**2**	**1**	**2**	**4**	**1**	**4**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**2**	**2**	**3**	**2**	**3**	**4**	**2**	**2**	**3**	**1**	**1**	**4**	**4**	**1**	**1**

EXERCISE-III (Analytical Questions) Master Your Understanding

1. On dissolving moderate amount of sodium metal in liquid NH_3 at low temperature, which one of the following does ***not*** occur
(1) Blue coloured solution is obtained.
(2) Na^+ ions are formed in the solution.
(3) Liquid NH_3 becomes good conductor of electricity.
(4) Liquid ammonia remains diamagnetic.

SB0086

2. NaOH is manufactured by electrolysis of brine solution. The products of the reaction are
(1) Cl_2 and H_2 (2) Cl_2 and Na–Hg
(3) Cl_2 and Na (4) Cl_2 and O_2

SB0088

3. Which alkali metal on flame test gives red violet colour
(1) Li (2) Cs
(3) Na (4) Rb

SB0091

4. In presence of iron, alkali metal react with liquid ammonia and form
(1) Metal mixture + H_2
(2) Iron metal mixture + H_2
(3) Metal mixture
(4) Metal amide + H_2

SB0092

5. Photoelectric effect is maximum in
(1) Cs (2) Na (3) K (4) Li

SB0093

6. When a standard solution of NaOH is left in air for a few hours,
(1) a precipitate will form
(2) strength will decrease
(3) strength will increase
(4) the concentration of Na^+ ions remain same
(5) all are wrong

SB0094

7. Which is used in purification of air in the space craft.
(1) Slaked lime
(2) Quick lime
(3) Potassium superoxide
(4) $CaCl_2$

SB0095

8. Potassium superoxide is used in oxygen cylinders of space craft as it -
(1) Absorbs O_2
(2) Eliminate moisture
(3) Absorbs CO_2 and increases O_2 content
(4) Forms ozone

SB0097

9. Compounds of alkaline earth metals are generally less soluble in water than that of alkali metals because of :-
(1) High hydration energy
(2) More covalent character
(3) More ionic character
(4) Less lattice energy

SB0098

10. Common product of hydrolysis of MO_2 and M_2O_2 type oxides
(1) H_2O_2, O_2 (2) H_2O_2, OH^-
(3) H_2, O_2 (4) H_2, OH^-

SB0131

11. From $Mg(OH)_2$ to $Ba(OH)_2$ solubility, thermal stability and basic character
(1) increase, decrease, increase
(2) increase, increase, increase
(3) decrease, decrease, decrease
(4) increase, increase, decrease

SB0132

12. Dark blue solution of alkali metal in liquid ammonia produce _____ on evaporation :-
(1) Metal oxide
(2) Metal
(3) $[M(NH_3)_6]^+$
(4) Hexa ammoniated metal

SB0133

13. Which of the following used in manufacturing of Borax.
(1) $MgCl_2$ (2) Na_2CO_3
(3) Li_2CO_3 (4) None

SB0134

14. Low solubility of LiF is due to
(1) High hydration enthalpy
(2) High lattice enthalpy
(3) High melting point
(4) High boiling point

SB0135

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15. $CaCl_2$, $MgCl_2$ are impurities in NaCl due to their ____ nature.

(1) Deliquescent (2) Effloroscent

(3) Inert (4) All

SB0136

16. Photoelectric cells mainly contain which of the following alkali metal.

(1) Na (2) Cs

(3) Li (4) Be

SB0137

17. KO_2 is paramagnetic as it contains one unpaired electron in :-

(1) $\pi 2p$ (2) $\sigma 2p$

(3) $\delta 2s$ (4) $\pi^* 2p$

SB0138

18. Color of alkali metal + liq. NH_3 solution in less concentrated and more concentrated forms are :

(1) Bronze, Green (2) Blue, Bronze

(3) Blue, Green (4) Bronze, Pink

SB0139

19. For a good quality cement the ratio of lime to total oxides of Si, Al, Fe is close to

(1) 1.5 (2) 2.5

(3) 1.0 (4) 2.0

SB0140

20. Gypsum is added to cement as

(1) It fastens the setting process

(2) It slows down the setting process

(3) It increases the oxide content

(4) None of these

SB0141

21. Which of the following alkali metals chloride form hydrate

(1) LiCl (2) NaCl

(3) RbCl (4) KCl

SB0142

22. Which of the following will not form ethynide on reaction with ethyne

(1) K (2) Na

(3) Cs (4) Li

SB0143

23. Which of the following hydroxide will decompose on heating

(1) NaOH (2) KOH

(3) LiOH (4) CsOH

SB0144

24. Which of the following not an alkaline earth metal

(1) Be (2) Mg

(3) Ca (4) Sr

SB0145

25. Be and Mg do not impart any colour to Bunsen flame becuase

(1) they are very hard

(2) exhibit diagonal relation with Al and Li respectively

(3) of their high ionisation potential

(4) density of these elements are high

SB0146

26. Hydrated halide of which of the following metal suffer hydrolysis on heating

(1) Be (2) Mg

(3) Ca (4) 1 and 2 both

SB0147

27. Berylium hydroxide dissolves in excess of alkali to give

(1) $\left[Be(OH)_3\right]^-$

(2) $\left[Be(OH)_4\right]^{-2}$

(3) $\left[Be(OH)_6\right]^{-4}$

(4) does not react with alkali because of its basic nature

SB0148

EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	4	4	1	2	3	3	2	2	2	4	2	2	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27			
Ans.	2	4	2	4	2	1	4	3	1	3	4	2			

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